Module 2: Genome Visualisation Tools

Bacterial Genomic Surveillance

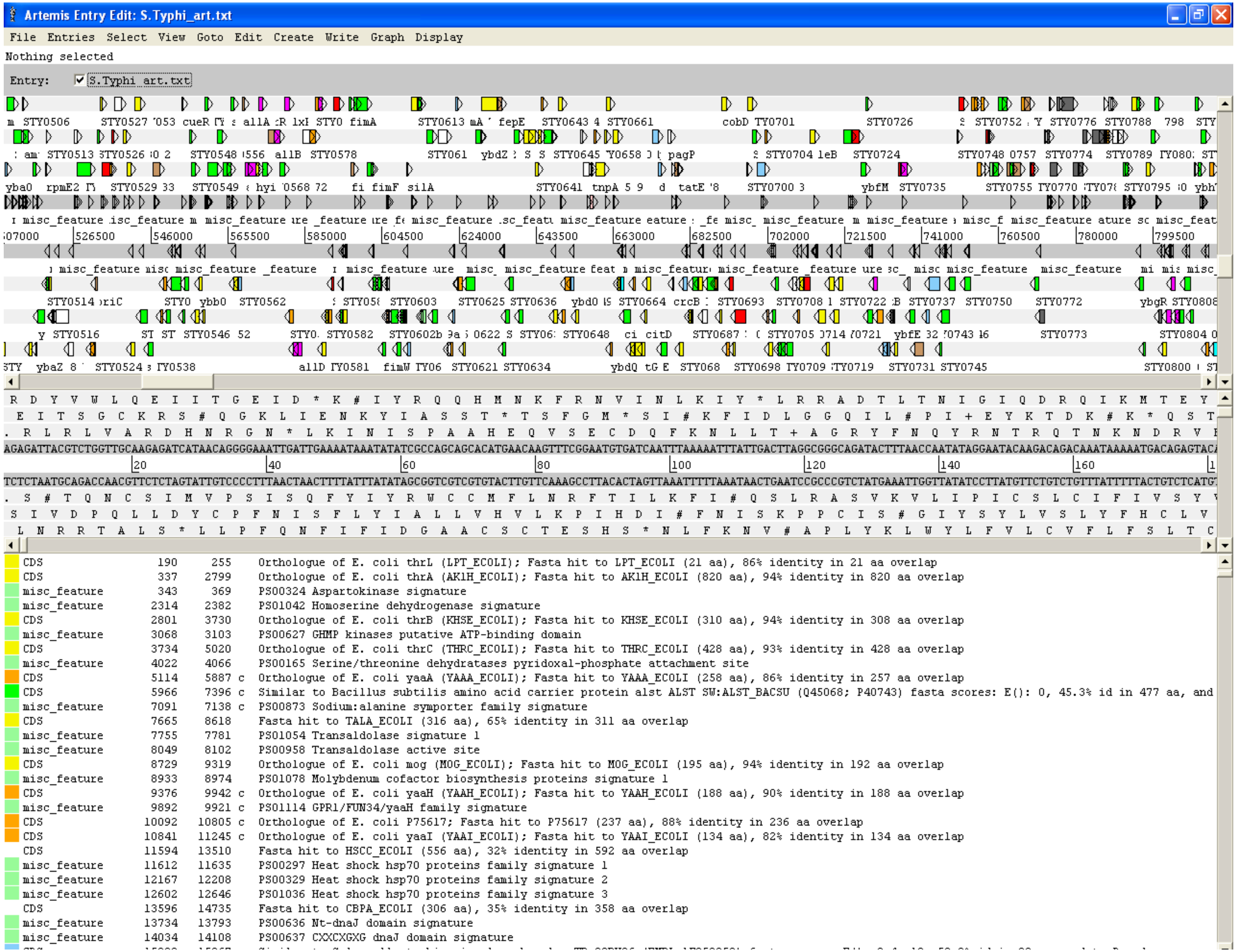
12, 22-26 June 2026
Department of Microbiology, Moyne Institute of Preventive Medicine, Trinity College Dublin, Ireland

Wellcome Connecting Science
Trinity College Dublin

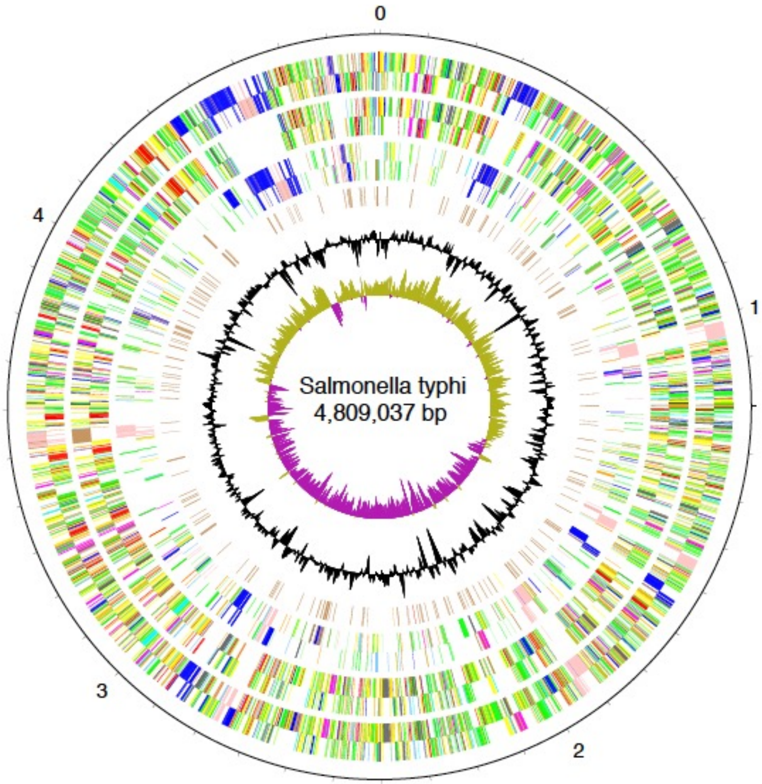
tcaattatac aagctcgaac ttcctgcctt ccctggggac gactgaacag ggatcctgat
gaataatgac atctgaaccc ggaaaacgcc gcaaaatcgc ctgctctacc tggtcagcca
caaaatgcgc ctgaacgagc ggcagattat cttccatttc caaatgaatc tgaataaagc
gagtcggccc tgactgccgc gtgcggagat cgtgcgcgcc gctgactccc ggccatgacg
tcacgatatc aataatttcc tgtcgttccg catcgggcag cgcgcgatcc aataatgact
gcaccgcctc ataccccata cgtaacgcgc tatacaaaat atagatgccg atacctaacg
caaacagggc atccgcacga tgccaaccgt accaggataa ccccagcgcg ataagaatcg
ctccgttcat cataacatca gactgataat gaagcatatc cgcccgtacc gcctgacttt
gggtcttgcg taccacccag cgctgaaacg tgaccagtat aatagtgcat atcagagcga
tgacggtaac gccaatcccc acgccggggt cgttcattgg cgtcggctta atcagattct
ggatactggt caaaaacaac gccgaaccgg aaataaacat actttgcgcc agcgccgcca
gcgactccgc tttaccgtga ccaaacgtat gttcatcatc tgcgggttgc agcgaataac
gcacgaccaa caaattcgtc aatgacgcgg caatatccac cagcgagtcc accaacgccg
ccagaatact caccgatccc gtataccacc acgcaaaaat tttgatcaaa agtaacgctg
atgccatcgc cgttgccgcg atagccgccc ggctaaccag ccgtccatag gtttgattca
taaatactcc cgctatcttc tgccggtagt ataacggatg gtttaaaaca gtaagctaat
aacagaatga caaactggag ataaaaaaaa cccccacatc atgtggggga agacagggat
ggtgtcacaa taaatcacct aacctgatga tataaatgga tttatatcaa ccgctgtcca
catagcgtcc acatcgttca gcgtccgttt aatcggacgc ttttaaattt ttatttttcc
cancagaaaa attagctttc ccgctgtttt gcaactcaac aagaggagat aaccgtcggt
aaacagccaa aagtatcttt aactgaacca gttcatctgc cccgatagt tgaaccaagt
ttttacgttg ttcttcacta agtgccatca cagattcaag tgatttaact gcgtcagaaa
caagttcaat ctctttaatt tgattctgta tacgcagctt ctctgatttc aactttgccc
attctctttg tcgatatgta cgcatcacct atcgtaaact ttgttggcag ccagaatact
tccaccaaca gcagtaacta actcagcaat ctgccaagag atatcaataa gagcagacaa
ttcaataaag ccaggagagg catactgaat ttttttcact acaggacgta aatctggtgg
agtcgcactg taagcccctc taaaaaagtt cactacactg tgcccaccct cccaaggcat
tgttgaatat cctcttgatc ctccttttag gccataaaaa aaaccatata gctgcaaata
ctctttcgtt aagaccgcga aatcagctaa ctcccaacga ttatcaagct caattataac
tttttccatt ttcaccaacc tgtccaactt taaaaattga cttttaagta tatgataaac
atggtttatt aaacgtgatc cactactttt aatgaagaat attccccaaa acagaaagcc
cctcgactga ggggctttct gtttgtaact acatccacat aatttgctgc cctgatggca
acgggtgcgg ccttacggcg tggacttctc ccggcttcac gatgtatcgc tgtaccgact
cataagtgat gaacgtggcg ctgcaattca cgttctggca ctggtgataa cgctcttttg
tcgtgtcagt gatatagcgg cttgtacgcg catgtgcggc atgctggcat aaaggacaat
gaaacatcgc gagcacctct tccggttttg ttgatggtgc cattttagtt aatttatcat
tataaaacaa acagataaga taaaacatca ctcatcatct tctgtttcgt actccacatc
agaaagcctg acctcaagct ctaaggacgt cgtgaagccg gaattattca gaaagtgtgt
caccttagtg attgtccagt cctgctcgtc tatgacgcgc ttaaagccag acactttgac
cggtgtttcc gtgtaaatat ctgcccgacc ggtagccagg ctgatggaga actccgcaac
gccccgttgc agtttatccc acttcgcctg agcggcgcgc atggcctgcg cttttgtggc
atataccgtg gtcagggcaa aaacgttgtc agcctcaccg gccatatatt caccttcgcg
cgcttcaggt acttttggcg ctttcttctg cgtgaccggt tttgctttcg ggtgctccag
tgcgcgcagg tgtttttctt tctttttgcg tttcagtttt accttctgct tttgtggttt
cgggtctttg gtgtgtaacc actttgccgt tacgccggtg taagctccac ggtcagcaat
cgcaaaatga tgacggtcgc cgtcgctgcg ggtgatggta atctgcggga tttttttacc
gctggccgtc accccctgcc ccgctttgag aaacaacagt tttcccattt ttaccgacac
ctcaccgccg ttgcgttctg caagacgggt cagaaatttc gcatcagact cctgcgactg
gtcgatgtgc aggattttaa ttccggccag tgacggagcg acactggctt ccagcctgtt
acgggaggct atcgcctcaa caatcgcacc gagcgtggtg tcatgccatg agccttcccg
gcgggaattg agcgtcccgc gaaaatctgc actccgggcg cggatggtga ccacatccgg
tgcgccccgg tgttcaacct catcaacggt aaatttccct ttgcatacca gggcaaaacc
tttccagccg atatacaccg tcaggacagc gccacgaacc ggcagcccga cctgcccgtc
ggcatcgttc agttcaatat caagctggtc agcctcaaag ccccggttat ccgtcagggt
catgctcatc agacggtcgc tgatattgcc ggtaatatcc ctgctgtcga gcatcagcat

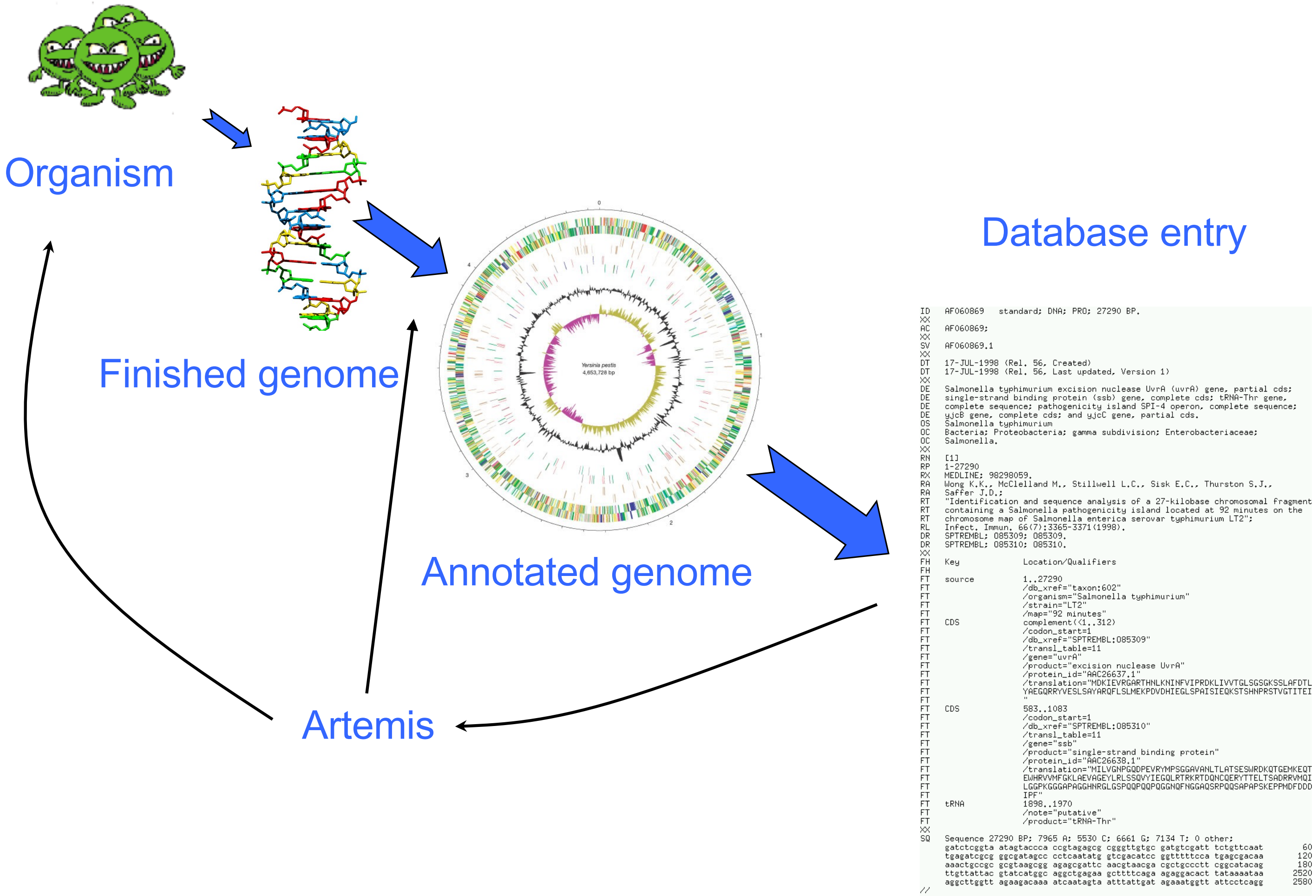
tcaattatac aagctcgaac ttcctgcctt ccctggggac gactgaacag ggatcctgat
gaataatgac atctgaaccc ggaaaacgcc gcaaaatcgc ctgctctacc tggtcagcca
caaaatgcgc ctgaacgagc ggcagattat cttccatttc caaatgaatc tgaataaagc
gagtcggccc tgactgccgc gtgcggagat cgtgcgcgcc gctgactccc ggccatgacg
tcacgatatc aataatttcc tgtcgttccg catcgggcag cgcgcgatcc aataatgact
gcaccgcctc ataccccata cgtaacgcgc tatacaaaat atagatgccg atacctaacg
caaacagggc atccgcacga tgccaaccgt accaggataa ccccagcgcg ataagaatcg
ctccgttcat cataacatca gactgataat gaagcatatc cgcccgtacc gcctgacttt
gggtcttgcg taccacccag cgctgaaacg tgaccagtat aatagtgcat atcagagcga
tgacggtaac gccaatcccc acgccggggt cgttcattgg cgtcggctta atcagattct
ggatactggt caaaaacaac gccgaaccgg aaataaacat actttgcgcc agcgccgcca
gcgactccgc tttaccgtga ccaaacgtat gttcatcatc tgcgggttgc agcgaataac
gcacgaccaa caaattcgtc aatgacgcgg caatatccac cagcgagtcc accaacgccg
ccagaatact caccgatccc gtataccacc acgcaaaaat tttgatcaaa agtaacgctg
atgccatcgc cgttgccgcg atagccgccc ggctaaccag ccgtccatag gtttgattca
taaatactcc cgctatcttc tgccggtagt ataacggatg gtttaaaaca gtaagctaat
aacagaatga caaactggag ataaaaaaaa cccccacatc atgtggggga agacagggat
ggtgtcacaa taaatcacct aacctgatga tataaatgga tttatatcaa ccgctgtcca
catagcgtcc acatcgttca gcgtccgttt aatcggacgc ttttaaattt ttatttttcc
cancagaaaa attagctttc ccgctgtttt gcaactcaac aagaggagat aaccgtcggt
aaacagccaa aagtatcttt aactgaacca gttcatctgc cccgatagt tgaaccaagt
ttttacgttg ttcttcacta agtgccatca cagattcaag tgatttaact gcgtcagaaa
caagttcaat ctctttaatt tgattctgta tacgcagctt ctctgatttc aactttgccc
attctctttg tcgatatgta cgcatcacct atcgtaaact ttgttggcag ccagaatact
tccaccaaca gcagtaacta actcagcaat ctgccaagag atatcaataa gagcagacaa
ttcaataaag ccaggagagg catactgaat ttttttcact acaggacgta aatctggtgg
agtcgcactg taagcccctc taaaaaagtt cactacactg tgcccaccct cccaaggcat
tgttgaatat cctcttgatc ctccttttag gccataaaaa aaaccatata gctgcaaata
ctctttcgtt aagaccgcga aatcagctaa ctcccaacga ttatcaagct caattataac
tttttccatt ttcaccaacc tgtccaactt taaaaattga cttttaagta tatgataaac
atggtttatt aaacgtgatc cactactttt aatgaagaat attccccaaa acagaaagcc
cctcgactga ggggctttct gtttgtaact acatccacat aatttgctgc cctgatggca
acgggtgcgg ccttacggcg tggacttctc ccggcttcac gatgtatcgc tgtaccgact
cataagtgat gaacgtggcg ctgcaattca cgttctggca ctggtgataa cgctcttttg
tcgtgtcagt gatatagcgg cttgtacgcg catgtgcggc atgctggcat aaaggacaat
gaaacatcgc gagcacctct tccggttttg ttgatggtgc cattttagtt aatttatcat
tataaaacaa acagataaga taaaacatca ctcatcatct tctgtttcgt actccacatc
agaaagcctg acctcaagct ctaaggacgt cgtgaagccg gaattattca gaaagtgtgt
caccttagtg attgtccagt cctgctcgtc tatgacgcgc ttaaagccag acactttgac
cggtgtttcc gtgtaaatat ctgcccgacc ggtagccagg ctgatggaga actccgcaac
gccccgttgc agtttatccc acttcgcctg agcggcgcgc atggcctgcg cttttgtggc
atataccgtg gtcagggcaa aaacgttgtc agcctcaccg gccatatatt caccttcgcg
cgcttcaggt acttttggcg ctttcttctg cgtgaccggt tttgctttcg ggtgctccag
tgcgcgcagg tgtttttctt tctttttgcg tttcagtttt accttctgct tttgtggttt
cgggtctttg gtgtgtaacc actttgccgt tacgccggtg taagctccac ggtcagcaat
cgcaaaatga tgacggtcgc cgtcgctgcg ggtgatggta atctgcggga tttttttacc
gctggccgtc accccctgcc ccgctttgag aaacaacagt tttcccattt ttaccgacac
ctcaccgccg ttgcgttctg caagacgggt cagaaatttc gcatcagact cctgcgactg
gtcgatgtgc aggattttaa ttccggccag tgacggagcg acactggctt ccagcctgtt
acgggaggct atcgcctcaa caatcgcacc gagcgtggtg tcatgccatg agccttcccg
gcgggaattg agcgtcccgc gaaaatctgc actccgggcg cggatggtga ccacatccgg
tgcgccccgg tgttcaacct catcaacggt aaatttccct ttgcatacca gggcaaaacc
tttccagccg atatacaccg tcaggacagc gccacgaacc ggcagcccga cctgcccgtc
ggcatcgttc agttcaatat caagctggtc agcctcaaag ccccggttat ccgtcagggt
catgctcatc agacggtcgc tgatattgcc ggtaatatcc ctgctgtcga gcatcagcat

tcaattatac aagctcgaac ttcctgcctt ccctggg
gaataatgac atctgaaccc ggaaaacgcc gcaaaat
caaaatgcgc ctgaacgagc ggcagattat cttccat
gagtcggccc tgactgccgc gtgcggagat cgtgcgc
tcacgatatc aataatttcc tgtcgttccg catcggg
gcaccgcctc ataccccata cgtaacgcgc tatacaa
caaacagggc atccgcacga tgccaaccgt accagga
ctccgttcat cataacatca gactgataat gaagcat
gggtcttgcg taccacccag cgctgaaacg tgaccag
tgacggtaac gccaatcccc acgccggggt cgttcat
ggatactggt caaaaacaac gccgaaccgg aaataaa
gcgactccgc tttaccgtga ccaaacgtat gttcatc
gcacgaccaa caaattcgtc aatgacgcgg caatatc
ccagaatact caccgatccc gtataccacc acgcaaa
atgccatcgc cgttgccgcg atagccgccc ggctaac
taaatactcc cgctatcttc tgccggtagt ataacgg
aacagaatga caaactggag ataaaaaaaa cccccac
ggtgtcacaa taaatcacct aacctgatga tataaat
catagcgtcc acatcgttca gcgtccgttt aatcgga
cancagaaaa attagctttc ccgctgtttt gcaactc
aaacagccaa aagtatcttt aactgaacca gttcatc
ttttacgttg ttcttcacta agtgccatca cagattc
caagttcaat ctctttaatt tgattctgta tacgcag
attctctttg tcgatatgta cgcatcacct atcgtaa
tccaccaaca gcagtaacta actcagcaat ctgccaa
ttcaataaag ccaggagagg catactgaat ttttttc
agtcgcactg taagcccctc taaaaaagtt cactaca
tgttgaatat cctcttgatc ctccttttag gccataa
ctctttcgtt aagaccgcga aatcagctaa ctcccaa
tttttccatt ttcaccaacc tgtccaactt taaaaat
atggtttatt aaacgtgatc cactactttt aatgaag
cctcgactga ggggctttct gtttgtaact acatcca
acgggtgcgg ccttacggcg tggacttctc ccggctt
cataagtgat gaacgtggcg ctgcaattca cgttctg
tcgtgtcagt gatatagcgg cttgtacgcg catgtgc
gaaacatcgc gagcacctct tccggttttg ttgatgg
tataaaacaa acagataaga taaaacatca ctcatca
agaaagcctg acctcaagct ctaaggacgt cgtgaag
caccttagtg attgtccagt cctgctcgtc tatgacg
cggtgtttcc gtgtaaatat ctgcccgacc ggtagcc
gccccgttgc agtttatccc acttcgcctg agcggcg
atataccgtg gtcagggcaa aaacgttgtc agcctca
cgcttcaggt acttttggcg ctttcttctg cgtgacc
tgcgcgcagg tgtttttctt tctttttgcg tttcagt
cgggtctttg gtgtgtaacc actttgccgt tacgccg
cgcaaaatga tgacggtcgc cgtcgctgcg ggtgatg
gctggccgtc accccctgcc ccgctttgag aaacaac
ctcaccgccg ttgcgttctg caagacgggt cagaaat
gtcgatgtgc aggattttaa ttccggccag tgacgga
acgggaggct atcgcctcaa caatcgcacc gagcgtg
gcgggaattg agcgtcccgc gaaaatctgc actccgg
tgcgccccgg tgttcaacct catcaacggt aaatttc
tttccagccg atatacaccg tcaggacagc gccacga
ggcatcgttc agttcaatat caagctggtc agcctca
catgctcatc agacggtcgc tgatattgcc ggtaata



Artemis Entry Edit: S.Typhi_art.txt
File Entries Select View Goto Edit Create Write Graph Display
Nothing selected
Entry: S.Typhi_art.txt
CDS 190 255 Orthologue of E. coli thrL (LPT_ECOLI); Fasta hit to LPT_ECOLI (21 aa), 86% identity in 21 aa overlap
CDS 337 2799 Orthologue of E. coli thrA (AK1H_ECOLI); Fasta hit to AK1H_ECOLI (820 aa), 94% identity in 820 aa overlap
misc_feature 343 369 PS00324 Aspartokinase signature
misc_feature 2314 2382 PS01042 Homoserine dehydrogenase signature
CDS 2801 3730 Orthologue of E. coli thrB (KHSE_ECOLI); Fasta hit to KHSE_ECOLI (310 aa), 94% identity in 308 aa overlap
misc_feature 3068 3103 PS00627 GHMP kinases putative ATP-binding domain
CDS 3734 5020 Orthologue of E. coli thrC (THRC_ECOLI); Fasta hit to THRC_ECOLI (428 aa), 93% identity in 428 aa overlap
misc_feature 4022 4066 PS00165 Serine/threonine dehydratases pyridoxal-phosphate attachment site
CDS 5114 5887 c Orthologue of E. coli yaaA (YAAA_ECOLI); Fasta hit to YAAA_ECOLI (258 aa), 86% identity in 257 aa overlap
CDS 5966 7396 c Similar to Bacillus subtilis amino acid carrier protein alst ALST SW:ALST_BACSU (Q45068; P40743) fasta scores: E(): 0, 45.3% id in 477 aa, and
misc_feature 7091 7138 c PS00873 Sodium:alanine symporter family signature
CDS 7665 8618 Fasta hit to TALA_ECOLI (316 aa), 65% identity in 311 aa overlap
misc_feature 7755 7781 PS01054 Transaldolase signature 1
misc_feature 8049 8102 PS00958 Transaldolase active site
CDS 8729 9319 Orthologue of E. coli mog (MOG_ECOLI); Fasta hit to MOG_ECOLI (195 aa), 94% identity in 192 aa overlap
misc_feature 8933 8974 PS01078 Molybdenum cofactor biosynthesis proteins signature 1
CDS 9376 9942 c Orthologue of E. coli yaaH (YAAH_ECOLI); Fasta hit to YAAH_ECOLI (188 aa), 90% identity in 188 aa overlap
misc_feature 9892 9921 c PS01114 GPR1/FUN34/yaaH family signature
CDS 10092 10805 c Orthologue of E. coli P75617; Fasta hit to P75617 (237 aa), 88% identity in 236 aa overlap
CDS 10841 11245 c Orthologue of E. coli yaaI (YAAI_ECOLI); Fasta hit to YAAI_ECOLI (134 aa), 82% identity in 134 aa overlap
CDS 11594 13510 Fasta hit to HSCC_ECOLI (556 aa), 32% identity in 592 aa overlap
misc_feature 11612 11635 PS00297 Heat shock hsp70 proteins family signature 1
misc_feature 12167 12208 PS00329 Heat shock hsp70 proteins family signature 2
misc_feature 12602 12646 PS01036 Heat shock hsp70 proteins family signature 3
CDS 13596 14735 Fasta hit to CBPA_ECOLI (306 aa), 35% identity in 358 aa overlap
misc_feature 13734 13793 PS00636 Nt-dnaJ domain signature
misc_feature 14034 14108 PS00637 CXXCXGXG dnaJ domain signature


0
1
2
3
4
Salmonella typhi
4,809,037 bp



Organism
Finished genome
Annotated genome
Database entry
Artemis

The Artemis Software

The Artemis Software is a set of software tools for genome browsing and annotation

View on GitHub

Artemis

Artemis is a free genome browser and annotation tool that allows visualisation of sequence features, next generation data and the results of analyses within the context of the sequence, and also its six-frame translation. Artemis is written in Java, and is available for UNIX, Macintosh and Windows systems. It can read EMBL and GENBANK database entries or sequence in FASTA, indexed FASTA or raw format. Other sequence features can be in EMBL, GENBANK or GFF format. Full information about the latest release of Artemis can be found in the Artemis manual.

Artemis

- Free and simple to download
- Written in Java
- Available for Windows, Mac, Unix (Linux)
- Reads EMBL, Genbank/DDBJ format files.
- Also reads DNA sequence files in Fasta and “raw” format (but not Word!)

EMBL format file

```
ID   AF060869   standard; DNA; PRO; 27290 BP.
XX
AC   AF060869;
XX
SV   AF060869.1
XX
DT   17-JUL-1998 (Rel. 56, Created)
DT   17-JUL-1998 (Rel. 56, Last updated, Version 1)
XX
DE   Salmonella typhimurium excision nuclease UvrA (uvrA) gene, partial cds;
DE   single-strand binding protein (ssb) gene, complete cds; tRNA-Thr gene,
DE   complete sequence; pathogenicity island SPI-4 operon, complete sequence;
DE   yjcB gene, complete cds; and yjcC gene, partial cds.
OS   Salmonella typhimurium
OC   Bacteria; Proteobacteria; gamma subdivision; Enterobacteriaceae;
OC   Salmonella.
XX
RN   [1]
RP   1-27290
RX   MEDLINE; 98298059.
RA   Wong K.K., McClelland M., Stillwell L.C., Sisk E.C., Thurston S.J.,
RA   Saffer J.D.;
RT   "Identification and sequence analysis of a 27-kilobase chromosomal fragment
RT   containing a Salmonella pathogenicity island located at 92 minutes on the
RT   chromosome map of Salmonella enterica serovar typhimurium LT2";
RL   Infect. Immun. 66(7):3365-3371(1998).
DR   SPTREMBL; O85309; O85309.
DR   SPTREMBL; O85310; O85310.
XX
FH   Key             Location/Qualifiers
FH
FT   source          1..27290
FT                   /db_xref="taxon:602"
FT                   /organism="Salmonella typhimurium"
FT                   /strain="LT2"
FT                   /map="92 minutes"
FT   CDS             complement(<1..312)
FT                   /codon_start=1
FT                   /db_xref="SPTREMBL:O85309"
FT                   /transl_table=11
FT                   /gene="uvrA"
FT                   /product="excision nuclease UvrA"
FT                   /protein_id="AAC26637.1"
FT                   /translation="MDKIEVRGARTHNLKNINFVIPRDKLIVVTGLSGSGKSSLAFDTL
FT                   YAEGQRRYVESLSAYARQFLSLMEKPDVDHIEGLSPAISIEQKSTSHNPRSTVGTITEI
FT                   "
FT   CDS             583..1083
FT                   /codon_start=1
FT                   /db_xref="SPTREMBL:O85310"
FT                   /transl_table=11
FT                   /gene="ssb"
FT                   /product="single-strand binding protein"
FT                   /protein_id="AAC26638.1"
FT                   /translation="MILVGNPGQDPEVRYMPSGGAVANLTLATSESWRDKQTGEMKEQT
FT                   EWHRVVMFGKLAEVAGEYLRLSSQVYIEGQLRTRKRTDQNCQERYTTELTSADRRVMQI
FT                   LGGPKGGGAPAGGHNRGLGSPQQPQQPQGGNQFNGGAQSRPQQSAPAPSKEPPMDFDDD
FT                   IPF"
FT   tRNA            1898..1970
FT                   /note="putative"
FT                   /product="tRNA-Thr"
XX
SQ   Sequence 27290 BP; 7965 A; 5530 C; 6661 G; 7134 T; 0 other;
     gatctcggta atagtaccca ccgtagagcg cgggttgtgc gatgtcgatt tctgttcaat        60
     tgagatcgcg ggcgatagcc cctcaatatg gtcgacatcc ggtttttcca tgagcgacaa       120
     aaactgccgc gcgtaagcgg agagcgattc aacgtaacga cgctgccctt cggcatacag       180
     ttgttattac gtatcatggc aggctgagaa gcttttcaga agaggacact tataaaataa      2520
     aggcttggtt agaagacaaa atcaatagta atttattgat agaaatggtt attcctcagg      2580
//
```

EMBL Header

Annotation

Sequence

Key

Qualifier

Two-character line code indicates the type of information contained in the line

EMBL

```
ID   ECRSMA     standard; DNA; PRO; 500 BP.
XX
AC   L40173;
XX
SV   L40173.1
XX
DT   10-AUG-1995 (Rel. 44, Created)
DT   04-MAR-2000 (Rel. 63, Last updated, Version 4)
XX
DE   Erwinia carotovora repressor (rsmA) gene, complete cds.
XX
KW   repressor; rsmA gene.
XX
OS   Pectobacterium carotovorum
OC   Bacteria; Proteobacteria; Gammaproteobacteria; Enterobacteriaceae;
OC   Pectobacterium.
XX
RN   [1]
RP   1-500
RA   Cui Y., Chatterjee A., Liu Y., Dumenyo C.K., Chatterjee A.K.;
RT   "Identification of a global repressor gene, rsmA, of Erwinia carotovora
RT   subsp. carotovora that controls extracellular enzymes,
RT   N-(3-oxohexanoyl)-L-homoserine lactone, and pathogenicity in soft-rotting
RT   Erwinia spp";
RL   J. Bacteriol. 177(17):0-0(1995).
XX
DR   GOA; Q47620; Q47620.
DR   SWISS-PROT; Q47620; CSRA_ERWCA.
XX
FH   Key             Location/Qualifiers
FH
FT   source          1..500
FT                   /db_xref="taxon:554"
FT                   /organism="Pectobacterium carotovorum"
FT                   /strain="71"
FT                   /sub_species="carotovora"
FT   -10_signal      107..112
FT                   /gene="rsmA"
FT   RBS             235..239
FT                   /gene="rsmA"
FT   CDS             246..431
FT                   /codon_start=1
FT                   /db_xref="GOA:Q47620"
FT                   /db_xref="SWISS-PROT:Q47620"
FT                   /note="putative"
FT                   /transl_table=11
FT                   /gene="rsmA"
FT                   /function="global repressor"
FT                   /protein_id="AAA74502.1"
FT                   /translation="MLILTRRVGETLIIGDEVTVTVLGVKGNQVRIGVNAPKEVSVHRE
FT                   EIYQRIQAEKSQPTSY"
XX
SQ   Sequence 500 BP; 140 A; 101 C; 120 G; 139 T; 0 other;

 ggatccggca agcaggatag aaagtgtgtt accttcagat attctgaagc tttacatgct        60
     cagttctgtt gttgtgataa caaaagcaca agctactgat atcgactaaa ctaacaagta
//
```

Genbank

```
LOCUS       ERWRSMA                  500 bp    DNA     linear   BCT 19-AUG-1995
DEFINITION  Erwinia carotovora repressor (rsmA) gene, complete cds.
ACCESSION   L40173
VERSION     L40173.1  GI:927031
KEYWORDS    repressor; rsmA gene.
SOURCE      Pectobacterium carotovorum
  ORGANISM  Pectobacterium carotovorum
            Bacteria; Proteobacteria; Gammaproteobacteria; Enterobacteriaceae;
            Pectobacterium.
REFERENCE   1  (bases 1 to 500)
  AUTHORS   Cui,Y., Chatterjee,A., Liu,Y., Dumenyo,C.K. and Chatterjee,A.K.
  TITLE     Identification of a global repressor gene, rsmA, of Erwinia
            carotovora subsp. carotovora that controls extracellular enzymes,
            N-(3-oxohexanoyl)-L-homoserine lactone, and pathogenicity in
            soft-rotting Erwinia spp
  JOURNAL   J. Bacteriol. 177(17) (1995) In press
COMMENT     Original source text: Erwinia carotovora (strain 71, sub_species
            carotovora) DNA.
FEATURES             Location/Qualifiers
     source          1..500
                     /organism="Pectobacterium carotovorum"
                     /strain="71"
                     /sub_species="carotovora"
                     /db_xref="taxon:554"
     gene            107..431
                     /gene="rsmA"
     -10_signal      107..112
                     /gene="rsmA"
     RBS             235..239
                     /gene="rsmA"
     CDS             246..431
                     /gene="rsmA"
                     /function="global repressor"
                     /note="putative"
                     /codon_start=1
                     /transl_table=11
                     /protein_id="AAA74502.1"
                     /db_xref="GI:927032"
                     /translation="MLILTRRVGETLIIGDEVTVTVLGVKGNQVRIGVNAPKEVSVHR
                     EEIYQRIQAEKSQPTSY"
BASE COUNT      140 a    101 c    120 g    139 t
ORIGIN
        1 ggatccggca agcaggatag aaagtgtgtt accttcagat attctgaagc tttacatgct
       61 cagttctgtt gttgtgataa caaaagcaca agctactgat atcgactaaa ctaacaagta
      121 gtgacaaacc ggagtgtgat ggtgtggtta taccatcgtc taggtttacg ttttcacagc
      181 acatgatgga taatggcggg gagacagaga gacccgactc tttataatct ttcaaggagc
      241 aaagaatgct tattttgact cgtcgagttg gcgaaaccct catcatcggc gatgaggtaa
      301 cggttaccgt attaggagtg aaaggcaacc aggtgcgtat tggtgttaat gcacctaaag
      361 aggtttctgt ccaccgtgaa gagatctatc agcgtattca ggccgaaaaa tctcaaccaa
      421 cgtcatattg attgacaatg cgtctcgtgt tcgcgggacg caattgttat ttccggtttt
      481 tcccccacac atttatcgat
//
```

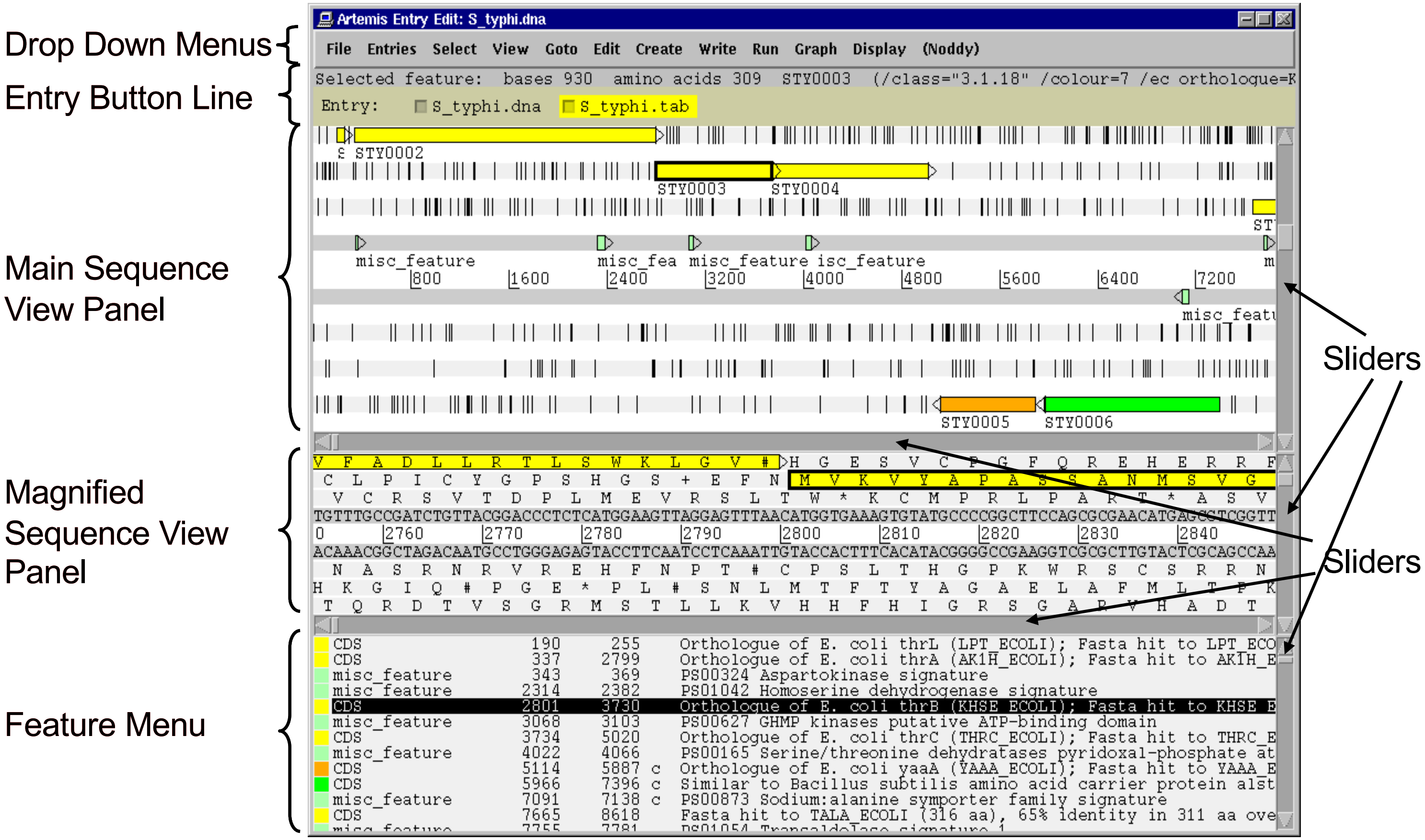
Artemis

- Sequence viewer and analysis tool

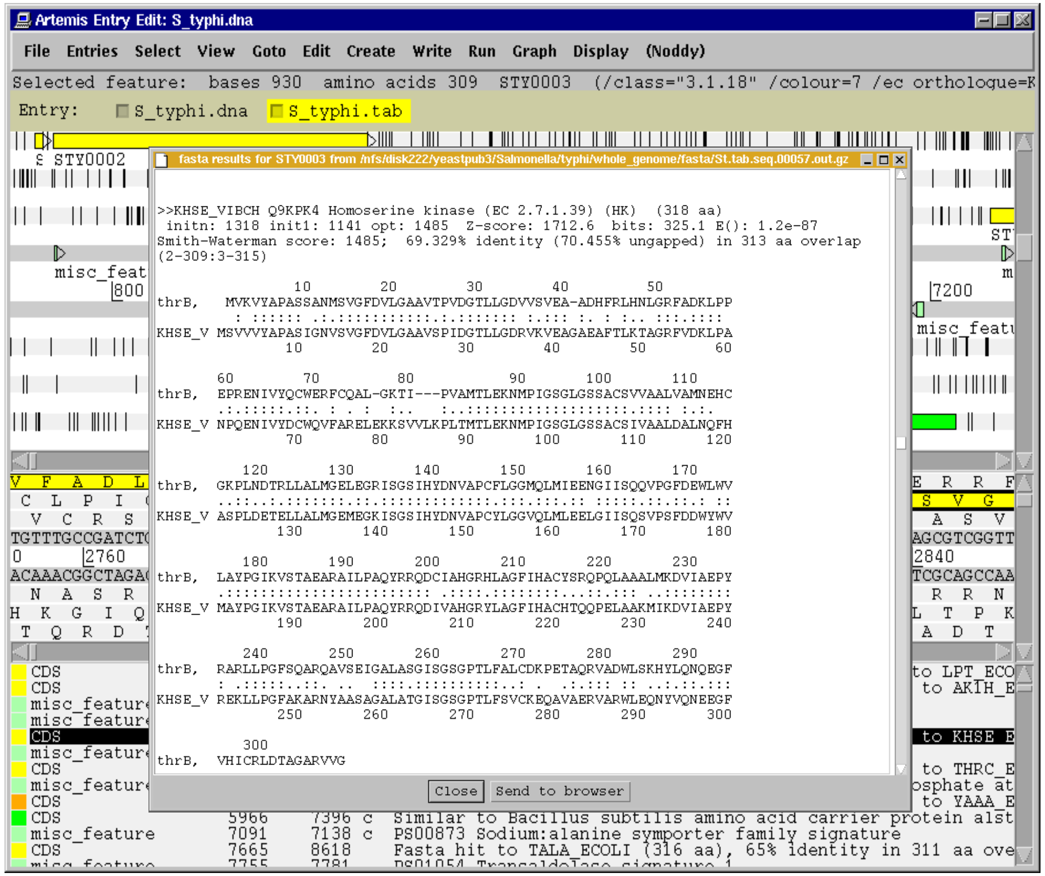
  - Visualization of sequence features
    - DNA
    - Six frame translation

  - Perform and view analysis
    - Basic analysis
    - Launch more complex analysis and searches
    - Import and view the results of other searches

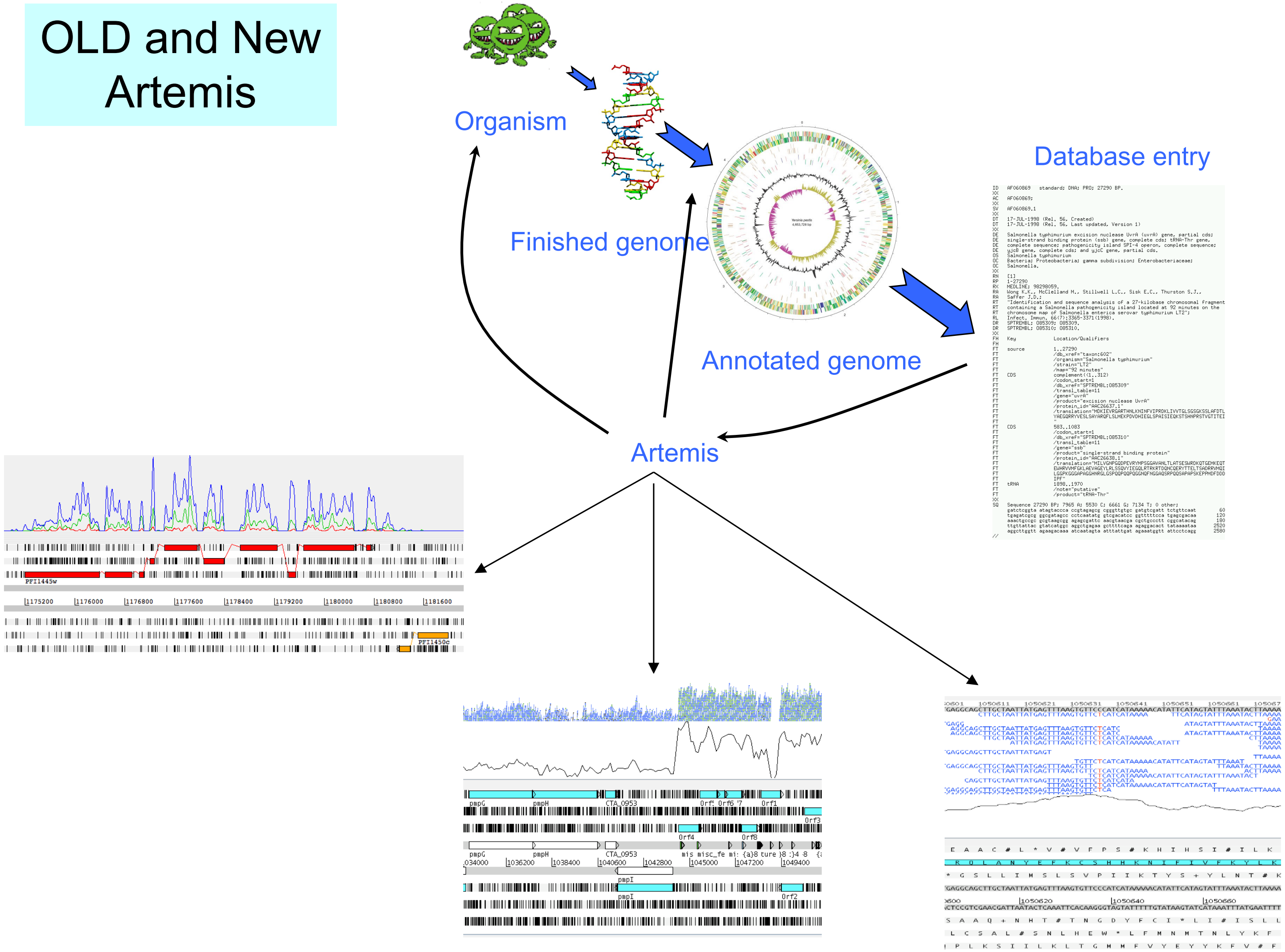
Artemis


Drop Down Menus
Entry Button Line
Main Sequence View Panel
Magnified Sequence View Panel
Feature Menu
Sliders
Sliders
Artemis Entry Edit: S_typhi.dna
File Entries Select View Goto Edit Create Write Run Graph Display (Noddy)
Selected feature: bases 930 amino acids 309 STY0003 (/class="3.1.18" /colour=7 /ec orthologue=K
Entry: S_typhi.dna S_typhi.tab
STY0002
STY0003 STY0004
misc_feature misc_fea misc_feature isc_feature
800 1600 2400 3200 4000 4800 5600 6400 7200
misc_featu
STY0005 STY0006
CDS 190 255 Orthologue of E. coli thrL (LPT_ECOLI); Fasta hit to LPT_ECO
CDS 337 2799 Orthologue of E. coli thrA (AK1H_ECOLI); Fasta hit to AK1H_E
misc_feature 343 369 PS00324 Aspartokinase signature
misc_feature 2314 2382 PS01042 Homoserine dehydrogenase signature
CDS 2801 3730 Orthologue of E. coli thrB (KHSE_ECOLI); Fasta hit to KHSE_E
misc_feature 3068 3103 PS00627 GHMP kinases putative ATP-binding domain
CDS 3734 5020 Orthologue of E. coli thrC (THRC_ECOLI); Fasta hit to THRC_E
misc_feature 4022 4066 PS00165 Serine/threonine dehydratases pyridoxal-phosphate at
CDS 5114 5887 c Orthologue of E. coli yaaA (YAAA_ECOLI); Fasta hit to YAAA_E
CDS 5966 7396 c Similar to Bacillus subtilis amino acid carrier protein alst
misc_feature 7091 7138 c PS00873 Sodium:alanine symporter family signature
CDS 7665 8618 Fasta hit to TALA_ECOLI (316 aa), 65% identity in 311 aa ove

Artemis


Artemis Entry Edit: S_typhi.dna
File Entries Select View Goto Edit Create Write Run Graph Display (Noddy)
Selected feature: bases 930 amino acids 309 STY0003 (/class="3.1.18" /colour=7 /ec orthologue=K
Entry: S_typhi.dna S_typhi.tab
fasta results for STY0003 from /nfs/disk222/yeastpub3/Salmonella/typhi/whole_genome/fasta/St.tab.seq.00057.out.gz
>>KHSE_VIBCH Q9KPK4 Homoserine kinase (EC 2.7.1.39) (HK) (318 aa)
initn: 1318 init1: 1141 opt: 1485 Z-score: 1712.6 bits: 325.1 E(): 1.2e-87
Smith-Waterman score: 1485; 69.329% identity (70.455% ungapped) in 313 aa overlap
(2-309:3-315)
Close
Send to browser

OLD and New Artemis



Outline of Artemis activities

- Artemis window features
- Open a genome sequence
- Changing the view
- Getting around
  - Goto Menu
  - Navigator
  - Feature Selector
- Basic analysis
  - Edit a feature
  - Show feature plots